Bioenergy

Bioenergy is generated by burning biomass—organic material including plant waste and animal matter—in power stations or by converting by-products into biofuels. Biomass is considered a renewable energy source because harvested crops and trees can be replaced. However, expanding the scale of bioenergy is problematic because it would require the conversion of arable land used for food production.

Sewage
Sludge from sewage treatment is broken down by anaerobic bacteria in digester tanks to produce methane and other gases, which can be purified and burned as fuel.

Industrial residue
Certain wastes left over from industrial processes—particularly black liquor from wood pulp and paper production—are rich in organic matter and can be burned as fuel to power electricity generators.

Agricultural
Crops, including rapeseed, sugar cane, and beet, are grown for processing into biofuels. Nonfood energy crops are sometimes grown on land that would otherwise have little agricultural value.

Forestry
Trees are the most ancient of fuels, burned for heat and light for thousands of years. Logs, chips, wood pellets, and sawdust account for over a third of all biomass energy used.

Animal waste
While animal remains can be burned as biomass, the manure produced by farmed animals, including cows, can also be treated to produce a methane-rich "biogas" that can be burned.

Municipal solid waste
Some of the vast amounts of solid waste produced is burned to generate heat or electricity. This has the added benefit of reducing the amount of space needed for landfill sites.

ETHANOL BIOFUEL

Ethanol is an alcohol that is produced from the sugars found in biomass crops, including sugar cane, corn, and sorghum. In Brazil, the world's leading producer of biofuel ethanol, more than 80 percent of all new cars and almost half of all motorcycles are able to run on ethanol or a gasoline-ethanol blend.

